

Finding the Equation of a Line

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Find the y -value at $x = 0$, then find the Gradient amount between y values.



Example

x	0	1	2	3	4
y	5	7	9	11	13

At $x = 0$
 $y = +5$

Gradient: $+2$

1

x	0	1	2	3	4
y	-2	1	4	7	10

At $x = 0$
 $y = \square$

Gradient: $\square$

2

x	0	1	2	3	4
y	0	5	10	15	20

At $x = 0$
 $y = \square$

Gradient: $\square$

3

x	0	1	2	3	4
y	-9	-10	-11	-12	-13

At $x = 0$
 $y = \square$

Gradient: $\square$

4

x	-2	-1	0	1	2
y	5	6	7	8	9

At $x = 0$
 $y = \square$

Gradient: $\square$

5

x	-2	-1	0	1	2
y	-2	-1	0	1	2

At $x = 0$
 $y = \square$

Gradient: $\square$

Now find the equations of the lines given the table of values.

Example

x	0	1	2	3	4
y	-1	-3	-5	-7	-9

At $x = 0$
 $y = -1$

Gradient: -2

The y -value goes at the back, including the sign.

Gradient goes in front of the x including the sign.

Equation: $y = -2x - 1$

Answer these using your Column 1 answers.

6 Equation: $y = \square x \square \square$

For Q.6 Use Gradient and y -value from Q.1

7 Equation: $y = \square x$

For Q.7 Use Gradient and y -value from Q.2

8 Equation: $y = \square x \square \square$

For Q.8 Use Gradient and y -value from Q.3

9 Equation: $y = \square$

For Q.9 Use Gradient and y -value from Q.4

10 Equation: $y = \square$

For Q.10 Use Gradient and y -value from Q.5

11

x	-3	-2	-1	0	1
y	32	22	12	2	-8

At $x = 0$
 $y = \square$

Gradient: $\square$

Equation: $y = \square$

12

x	0	1	2	3	4
y	8	10	12	14	16

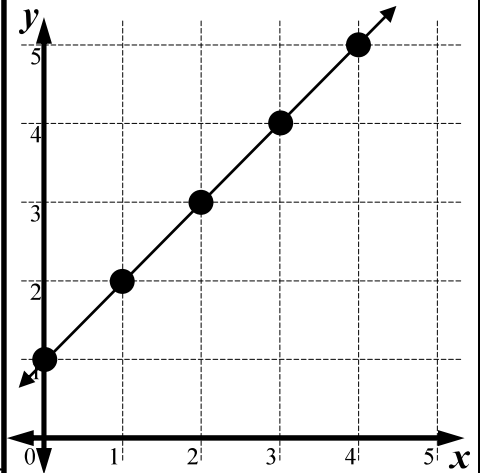
At $x = 0$
 $y = \square$

Gradient: $\square$

Equation: $y = \square$

Change the points on the graph to a table then find the line equation by the same method.

13



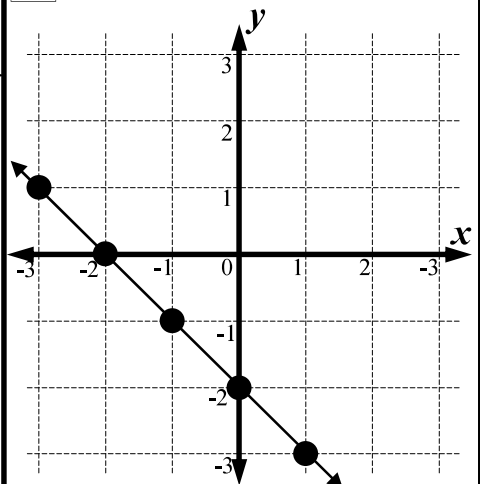
x				
y				

At $x = 0$
 $y = \square$

Gradient: $\square$

Equation: $y = \square$

14



x				
y				

At $x = 0$
 $y = \square$

Gradient: $\square$

Equation: $y = \square$

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Find the y -value at $x = 0$, then find the Gradient amount between y values.



Example

x	0	1	2	3	4
y	5	7	9	11	13

At $x = 0$

$y = +5$

Gradient: $+2$

1

x	0	1	2	3	4
y	-2	1	4	7	10

At $x = 0$

$y = -2$

Gradient: $+3$

2

x	0	1	2	3	4
y	0	5	10	15	20

At $x = 0$

$y = 0$

Gradient: $+5$

3

x	0	1	2	3	4
y	-9	-10	-11	-12	-13

At $x = 0$

$y = -9$

Gradient: -1

4

x	-2	-1	0	1	2
y	5	6	7	8	9

At $x = 0$

$y = 7$

Gradient: $+1$

5

x	-2	-1	0	1	2
y	-2	-1	0	1	2

At $x = 0$

$y = 0$

Gradient: $+1$

Now find the equations of the lines given the table of values.

Example

x	0	1	2	3	4
y	-1	-3	-5	-7	-9

At $x = 0$

$y = -1$

Gradient: -2

The y -value goes at the back, including the sign.

Gradient goes in front of the x including the sign.

Equation: $y = -2x - 1$

Answer these using your Column 1 answers.

6 Equation: $y = 3x - 2$

For Q.6 Use Gradient and y -value from Q.1

7 Equation: $y = 5x$

For Q.7 Use Gradient and y -value from Q.2

8 Equation: $y = -x + 9$

For Q.8 Use Gradient and y -value from Q.3

9 Equation: $y = x + 7$

For Q.9 Use Gradient and y -value from Q.4

10 Equation: $y = x$

For Q.10 Use Gradient and y -value from Q.5

11

x	-3	-2	-1	0	1
y	32	22	12	2	-8

At $x = 0$

$y = 2$

Gradient: -10

Equation: $y = -10x + 2$

12

x	0	1	2	3	4
y	8	10	12	14	16

At $x = 0$

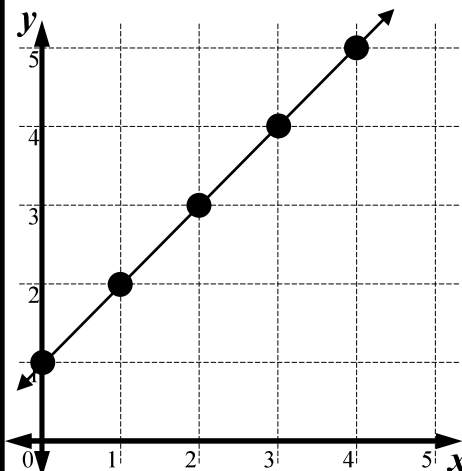
$y = 8$

Gradient: $+2$

Equation: $y = 2x + 8$

Change the points on the graph to a table then find the line equation by the same method.

13



x	0	1	2	3	4
y	1	2	3	4	5

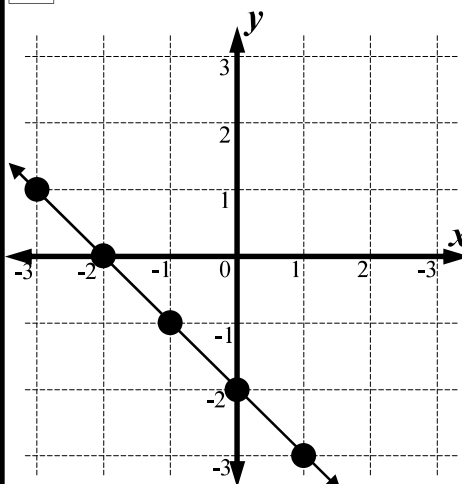
At $x = 0$

$y = 1$

Gradient: $+1$

Equation: $y = x + 1$

14



x	-3	-2	-1	0	1
y	1	0	-1	-2	-3

At $x = 0$

$y = -2$

Gradient: -1

Equation: $y = -x - 2$